

IEM KOLKATA**Calculus
Derivatives**

Q1) $f(x) = |x| = \begin{cases} -x & x < 0 \\ x & x \geq 0 \end{cases}$

Is $f(x)$ differentiable at $x = 0$.

Q2) Is $y = |2 - 3x|$ continuous & differentiable at $x = 2/3$?

Q3) Consider the function $f(x) = x^2 e^{-x}$, maximum occurs at $x =$?
A. 0 B. 1 C. 2 D. Cannot be determined

Q4) The maximum value of $f(x) = x^2 - x - 2$ in the interval $[-4, 4]$ is?
A. 18 B. 10 C. -2.25 D. indeterminate

Q5) The maximum value of $f(x) = x^3 - 9x^2 + 24x + 5$ in the interval $[1, 6]$ is equal to?
A. 21 B. 25 C. 41 D. 45

Q6) The number of distinct extrema for the curve $3x^4 - 16x^3 + 24x^2 + 37$ is?
A. 0 B. 1 C. 2 D. ∞

Q7) Which of the following conditions are required for the standard Mean Value Theorem on $[a, b]$?
A. f is continuous on $[a, b]$ and differentiable on (a, b) .
B. f is differentiable on $[a, b]$.
C. f is continuous only on (a, b) .
D. $f'(x) \neq 0$ throughout (a, b) .

Q8) The value of c of Mean Value Theorem for $f(x) = x^2 - 5x + 6$ in the interval $[2, 3]$ is?

A. $2/5$ B. $5/2$ C. 0 D. None of the above

Q9) The value of c of Mean Value Theorem for $f(x) = 1 + x^2$ in the interval $[1, 2]$ is?

A. $2/3$ B. $3/2$ C. 0 D. None of the above

Q10) Let f be continuous on $[1, 4]$, differentiable on $(1, 4)$, and satisfy $f(1) = 2$, $f(4) = 11$. Then there exists at least one $c \in (1, 4)$ such that $f'(c) = ?$

A. 2

B. 3

C. 4

D. 9

Q11) Consider $f(x) = x^3$ on $(-1, 2]$. The point(s) satisfying the Mean Value Theorem equation are

A. $c = 1$

B. $c = -1/\sqrt{3}$

C. $c = 2/\sqrt{3}$

D. $c = 1/\sqrt{3}$ is the only point in the interval.

Q12) The function $f(x) = x + 1/x$, $x > 0$, has its minimum value at

A. $x = 1$

B. $x = 1/2$

C. $x = 2$

D. $x = \sqrt{2}$

Q13) The maximum value of $f(x) = x(4 - x^2)$, $0 \leq x \leq 2$, is

A. -1

B. $16/(3\sqrt{3})$

C. $(-16)/(3\sqrt{3})$

D. $(-8)/(3\sqrt{3})$

Q14) A rectangular sheet has perimeter 20 units. If its length is x and breadth is $10 - x$, the maximum possible area is

- A. 20
- B. 25
- C. 50
- D. 100

Q15) Let $f(x) = x/(1 + x^2)$. The maximum value of $f(x)$ over all real x is

- A. 1
- B. $1/2$
- C. $1/4$
- D. 2